DAY 2

SPOT 1

1. Mechanics of respiration (pressure changes during inspiration and expiration)
2. Alveolar pressure becomes slightly negative (≈ –1 cm H_2O) during inspiration, allowing air entry
3. Contraction of the diaphragm
4. Due to elastic recoil of lungs pulling inward and chest wall pulling outward, creating sub-atmospheric pressure in the pleural space

SPOT 2

1. Alveolus (showing effect of surfactant on alveolar stability)
2. Dipalmitoyl phosphatidylcholine (DPPC) — a phospholipid
3. Reduces surface tension, preventing alveolar collapse at low lung volumes
4. Causes decreased lung compliance and alveolar collapse → Neonatal Respiratory Distress Syndrome (NRDS)

SPOT 3

1. Pressure–Volume (Compliance) curve of the lungs and thoracic wall
2. Change in lung volume per unit change in transpulmonary pressure
   $C = \Delta V / \Delta P$
3. Pulmonary fibrosis, pulmonary edema, or surfactant deficiency (any one)
4. Emphysema or aging (any one)

SPOT 4

1. Zones of pulmonary blood flow (West's Zones of the lung)
2. Gravity (hydrostatic pressure gradient)
3. Zone 3 (base of the lung)
4. Alveolar pressure (PA > Pa > Pv)

SPOT 5

1. Oxygen–Hemoglobin Dissociation Curve
2. Represents high PO_2 range where hemoglobin is almost fully saturated (facilitates O_2 loading in lungs)

3. ↑ Temperature, ↑ PCO_2, ↑ H^+ (↓ pH), ↑ 2,3-BPG

4. Promotes O_2 unloading in tissues with small fall in PO_2